METRIC

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SUPERSEDING
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31 March 1989

DETAIL SPECIFICATION

PROJECTILE, M940 – 20MM, MULTIPURPOSE, TRACE SELF-DESTRUCT, METAL PARTS

Reactivated after 3 May 2006 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the metal parts configuration of the 20mm multipurpose, trace self destruct projectile M940 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product
MIL-STD-651	-	Visual Inspection Standards for 20MM Ammunition and Components

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-SS, Picatinny, New Jersey 07806-5000, or ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIT online database at http://assist.daps.dla.mil.

AMSC N/A FSC: 1305

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MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card

(Copies of federal and military specifications, standards and handbooks are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111 - 5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC) PUBLICATIONS

SCATP-20	-	Ammunition Ballistic Acceptance Test Methods, Test Procedures for 20mm Cartridges

(This publication is available from US Army ARDEC, AMSRD-AAR-QEM-F, Picatinny, NJ 07806-5000.)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC) DRAWINGS

12551767	Cartridge 20mm MPT-SD, M940
12551771	Disc, Closure Nozzle
12551773	Nose Cap
12551778	Seal
12551779	Disc, Metering
12551780	Projectile Body Assembly, Painted
12551781	Projectile Body Assembly
12551782	Body, Projectile
12551783	Band, Rotating
12551784	Non-Destructive Test Standard
11075203	Eddy Current Flaw Detector

(Copies of these drawings may be requested on line at Drawing-Request@pica.army.mil or from US Army ARDEC, AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E384	-	Standard Test Method for Microhardness of Materials
ASTM E18	-	Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B117	-	Standard Method of Salt Spray (fog) Testing

(Copies of this document are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or www.astm.org.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Parts and Subassemblies. Materials, parts and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 Banding.

3.3.1 Band fill. The rotating band blank shall fill the band seat to include a minimum 50% of the volume of the dovetail undercut.

3.3.2 Band hardness. The finished rotating band shall conform to the hardness requirements specified on drawing 12551781.

3.4 Soundness. Metal discontinuity in the projectile body determined by eddy current inspection shall be less than that represented by non-destructive test standard, see drawing 12551784.

3.5 Protective coating.

3.5.1 Phosphate coating. The pretreatment cleaning process shall be in accordance with the applicable requirements of drawing 12551780. The phosphate coating shall be in accordance with the requirements of TT-C-490 for appearance and minimum weight as applicable to the type of phosphate coating used. Test specimens (panels) shall be in accordance with the requirements of TT-C-490.

3.5.2 Paint coating. The exterior coating of the projectile shall be in accordance with the applicable requirements of drawing 12551780. The exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490.

3.5.3 Corrosion resistance. The exterior painted surface of the projectile shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.6 Metal parts security. The projectile metal parts when assembled into test cartridges in accordance with drawing 12551767 (inert loaded projectile) shall not separate or break up during testing for metal parts security.

3.7 Crimp ability. The crimp lip of the projectile body shall be capable of being crimped, sufficient to retain metering disc during metal parts security testing, without exhibiting cracking anywhere on the periphery of the crimped surface.

3.8 Hardness.

3.8.1 Nose cap. The hardness shall be measured using a superficial hardness tester with direct reading on the Rockwell 15-T scale. The average of four readings shall determine hardness. No individual reading shall exceed the acceptable range (35-55) by more than 5 points R15-T.

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3.8.2 Projectile body. The projectile body shall comply with the hardness requirements of drawing 12551782.

3.9 Workmanship. All parts and assemblies shall be fabricated, loaded, and assembled in a thorough workmanlike manner. They shall be clean and free of burrs, sharp edges, cracks, scratches, dents, folds, wrinkles, buckles, dirt, grease, oil, rust, and other foreign matter. Exterior surface coatings shall be continuous; however, light scratches not exposing base material may be permitted.

4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix

METHOD OF VERIFICATION
N/A - Not applicable
1 - Analysis
2 - Demonstration
3 - Examination
4 - Test

CLASSES OF VERIFICATION
A - First article
B - Conformance

Section 3 Requirement		Verification Methods					Verification Class		Section 4 Method
		N/A	1	2	3	4	A	B	
3.1	First article				X	X	X		4.2
3.2	Parts and subassemblies				X		X	X	Table IV
3.3	Banding				X		X	X	4.4.1
3.4	Soundness					X	X	X	4.4.2
3.5	Protective coating					X	X	X	4.4.3
3.6	Metal parts security					X	X	X	4.4.4
3.7	Crimp ability				X		X	X	4.4.5
3.8	Hardness					X	X	X	4.4.6
3.9	Workmanship				X		X	X	Table IV

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When specified, a sample of 4045 M940 projectiles and projectile components as identified in Table II shall be subjected to first article verification inspections and tests with quantities in accordance with Table II and Table IV.

4.2.1 First article rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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TABLE II. First article inspection

Examination or Test 1/	Conformance Criteria		Requirement Paragraph	Inspection Method	Defect Classification
	Sample	Acc/Rej			
Examination for defects					
Rotating band	32	0/1	3.2/3.9	Table IV	Table IV
Projectile body (unbanded)	125	0/1			
Projectile body assembly (unpainted)	125	0/1			
Projectile body assembly, painted	125	0/1			
Disc closure nozzle	125	0/1			
Nose cap	125	0/1			
Seal	125	0/1			
Metering disc	125	0/1			
Rotating band					
Band fill (body and rotating band assembly, uncoated)	20	0/1	3.3.1	4.4.1.1	Major
Band hardness	20	0/1	3.3.2	4.4.1.2	Major
Soundness	4045	0/1	3.4	4.4.2	Major
Protective coating					
Phosphate coating (projectile metal parts before painting or standard panel) 3/	3	0/1	3.5.1	4.4.3.1	Major
Paint adhesion (projectile body assembly, painted)	13	0/1	3.5.2	4.4.3.2	Major
Corrosion resistance (projectile body assembly, painted)	32	0/1	3.5.3	4.4.3.3	Major
Metal parts security (projectile metal parts) 1/	4000	0/1	3.6	4.4.4	Major
Crimp ability	20	0/1	3.7	4.4.5	Major
Hardness					
Projectile Body	10	0/1	3.8.1	4.4.6.2	Major
Nose Cap	80	0/1	3.8.2	4.4.6.1	Major

See Notes after Table III

4.3 Conformance verification.

4.3.1 Conformance inspection. The sample cartridges shall be subjected to conformance verification in accordance with Table III and Table IV.

4.3.2 Classification of characteristics. Critical, major and minor characteristics are defined in MIL-STD-1916.

4.3.3 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. Lot numbering shall be in accordance with MIL-STD-1168.

4.3.4 Conformance rejection. If any sample fails to comply with the conformance inspection requirements, the lot shall be rejected.

4.3.5 Examinations and tests. The attribute sampling plan required for the examination for defects in Table IV shall be in accordance with the attribute sampling plan of MIL-STD-1916, using Verification Level IV for major characteristics and Level II for minor characteristics unless otherwise

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noted. One hundred percent inspection shall be used on all critical characteristics. The lot shall be suspended if a malfunction or casualty not covered by this specification occurs in any firing test (see 6.8).

4.3.6 Alternative conformance acceptance. Unless otherwise specified, alternate conformance procedures may be proposed (see 6.2).

TABLE III. Conformance inspection

Examination or Test 1/	Conformance Criteria		Requirement Paragraph	Inspection Method	Defect Classification
	Sample	Acc/Rej			
Examination for defects					
Rotating band	Table IV	0/1	3.2/3.9	Table IV	Table IV
Projectile body (unbanded)		0/1			
Projectile body assembly (unpainted)		0/1			
Projectile body assembly, painted		0/1			
Disc closure nozzle		0/1			
Nose cap		0/1			
Seal		0/1			
Metering disc		0/1			
Rotating band					
Band fill (body and rotating band assembly, uncoated)	20	see 2/	3.3.1	4.4.1.1	Major
Band hardness	20	see 2/	3.3.2	4.4.1.2	Major
Soundness	100%	see 5/	3.4	4.4.2	Major
Protective coating					
Phosphate coating (projectile metal parts before painting or standard panel) 3/	3	0/1	3.5.1	4.4.3.1	Major
Paint adhesion (projectile body assembly, painted)	13	see 2/	3.5.2	4.4.3.2	Major
Corrosion resistance (projectile body assembly, painted)	32	see 2/	3.5.3	4.4.3.3	Major
Metal parts security (projectile metal parts) 1/	250	0/1	3.6	4.4.4	Major
Crimp ability 4/	20	0/1	3.7	4.4.5	Major
Hardness					
Projectile Body	10	see 2/	3.8.1	4.4.6.2	Major
Nose Cap	80	see 2/	3.8.2	4.4.6.1	Major

Notes:

1/ The lot shall be rejected if any of the following types of metal parts separation or breakup occurs in the metal parts security test, as witnessed by holes, in the fragmentation screens, or recovery of the parts or fragments.

a. Body separation or breakup.

b. Loss of nose.

c. Complete rotating band separation. This is defined as detachment of the entire band having band sear knurling imprint.

d. Partial rotating band separation. This is defined as the detachment of part of the band having band sear knurling imprint.

e. The following criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint.

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1) The lot shall be rejected if three or more assemblies throw band fragments having chord lengths not greater than 0.500 inch.

2) The lot shall be rejected if any assembly throws a band fragment having a chord length greater than 0.500 inch (indicative of deep seated band lamination).

3) The lot shall not be penalized for band particles thrown as a result of normal band fringing or slivering.

2/ If two or more units of the sample fail to comply with the requirement, the lot shall be rejected. For lot conformance only, if one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If, in the accumulated samples, two or more units fail to comply, the lot shall be rejected.

3/ Failure of the phosphate coating on the sample projectiles or standard panels to comply with the applicable requirements shall cause rejection of all production projectile metal parts phosphated since the preceding test.

4/ If one or more of the crimped projectile bodies shows evidence of cracking on the periphery of the crimp the lot shall be rejected.

5/ A projectile that fails to comply with the soundness requirement shall be rejected.

TABLE IV. Examination for defects

Rotating Band, Dwg. 12551783				
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Critical	None defined	N/A	N/A	N/A
Major				
101	Foreign matter	Level IV	3.9	Visual
102	Defective rotating band metal, including lamination	Level IV	3.2	Visual
103	Incomplete or deformed 1/	Level IV	3.2	Visual
Minor				
201	0.095 width of rotating band	Level II	3.2	Gage
202	Poor workmanship	Level II	3.9	Visual
Projectile Body, Dwg. 12551782				
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Critical	None defined	N/A	N/A	N/A
Major				
101	Knurling missing or incomplete	Level IV	3.2	Visual
102	Foreign material in band seat	Level IV	3.9	Visual
103	0.012 min, depth of knurl 2/	Level IV	3.2	Visual
104	1.36 length, mouth to rear of rotating band	Level IV	3.2	Gage
105	0.784 diameter bourrelet, min	Level IV	3.2	Gage
106	0.769 diameter body rear of rotating band	Level IV	3.2	Gage
107	0.10 height of boss in tracer cavity	Level IV	3.2	Gage
108	0.62 depth metering disc to bottom of tracer cavity	Level IV	3.2	Gage
109	0.485 large diameter body cavity	Level IV	3.2	Gage
110	0.10 runout large body cavity	Level IV	3.2	Gage
111	1.69 depth body cavity	Level IV	3.2	Gage

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TABLE IV. Examination for defects – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Major				
112	0.423 small diameter body cavity	Level IV	3.2	Gage
113	0.005 runout small diameter body cavity	Level IV	3.2	Gage
114	0.087 length boss (tracer cavity to basic diameter (igniter cavity))	Level IV	3.2	Gage
115	0.035 thickness igniter boss base	Level IV	3.2	Gage
116	0.435 diameter tracer cavity	Level IV	3.2	Gage
117	0.008 runout tracer cavity	Level IV	3.2	Gage
118	0.515 diameter metering disc cavity	Level IV	3.2	Gage
119	0.12 depth metering disc cavity	Level IV	3.2	Gage
120	0.008 runout metering disc cavity	Level IV	3.2	Gage
121	0.517 diameter serration in mouth	Level IV	3.2	Gage
122	0.004 true position of serration in mouth	Level IV	3.2	Gage
123	0.563 diameter undercut in mouth	Level IV	3.2	Gage
124	0.456 depth mouth to undercut	Level IV	3.2	Gage
Minor				
201	0.680 diameter rotating band seat knurl	Level II	3.2	Gage
202	0.105 width rotating band seat	Level II	3.2	Visual/Gage
203	0.082 width knurling, minimum	Level II	3.2	Gage
204	0.645 diameter mouth outside projectile body	Level II	3.2	Gage
205	2.570 overall length of projectile body	Level II	3.2	Visual
206	250 micro-inch projectile body surface roughness	Level II	3.2	Visual
207	Poor workmanship	Level II	3.9	Gage
Projectile Body Assembly, Dwg. 12551781				
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Critical	None defined	N/A	N/A	N/A
Major				
101	Ferrous metal discontinuity (soundness)	100%	3.2	NDT
102	1.365 length mouth to rear edge of rotating band	Level IV	3.2	Gage
103	0.006 runout rotating band	Level IV	3.2	Gage
104	0.826 diameter of rotating band	Level IV	3.2	Gage
105	0.105 width rotating band	Level IV	3.2	Gage
106	Void at band/body interface	Level IV	3.2	Visual
107	0.19 location crimp groove	Level IV	3.2	Gage
108	0.074 radius, profile crimp groove	Level IV	3.2	Gage
109	0.005 max radius, rear edge of crimp groove	Level IV	3.2	Gage
Minor				
201	Poor workmanship	Level II	3.9	Visual
Projectile Body Assembly, Painted, Dwg. 12551780				
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Critical	None defined	N/A	N/A	N/A
Major				
101	0.786 diameter bourrelet, max	Level IV	3.2	Gage

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TABLE IV. Examination for defects – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Minor				
201	Exterior paint not dry	Level II	3.2/3.9	Taction
202	Marking incorrect, incomplete, illegible or missing	Level II	3.2/3.9	Visual
203	Paint on rotating band incomplete or missing	Level II	3.2/3.9	Visual
204	Interior paint coverage incomplete	Level II	3.2/3.9	Visual
205	Poor workmanship	Level II	3.9	Visual
Closure Nozzle Disc, Dwg. 12551771				
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Critical	None defined	N/A	N/A	N/A
Major				
101	0.114 through diameter of nozzle disc	Level IV	3.2	Gage
102	0.002 runout 0.409 small outside diameter to datums A and B of nozzle disc	Level IV	3.2	Gage
103	0.422 datum diameter of nozzle disc	Level IV	3.2	Gage
104	0.270 length datum B to face of nozzle disc	Level IV	3.2	Gage
105	0.04 length datum B to angle of nozzle disc	Level IV	3.2	Gage
106	63 micro-inch surface roughness of small diameter and angle surface of nozzle disc	Level IV	3.2	Visual
Minor				
201	0.010 runout, 0.24 counterbore, diameter to datum of nozzle disc	Level II	3.2	Gage
202	0.002 runout, 0.516 diameter to datums A and B of nozzle disc	Level II	3.2	Gage
203	Foreign matter	Level II	3.9	Visual
204	Poor workmanship	Level II	3.9	Visual
Nose Cap, Dwg. 12551773				
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Critical	None defined	N/A	N/A	N/A
Major				
101	0.025 nose tip thickness	Level IV	3.2	Gage
102	0.439 length of nose	Level IV	3.2	Gage
103	1.220 length, nose contour	Level IV	3.2	Gage
104	Outside nose contour	Level IV	3.2	Gage
105	0.047 spherical nose radius	Level IV	3.2	Gage
106	0.222 nose location first inside diameter from datum	Level IV	3.2	Gage
107	0.333 location inside diameter from inside nose tip	Level IV	3.2	Gage
108	0.412 diameter, large inside cylindrical, nose cap	Level IV	3.2	Gage
109	0.368 diameter, small inside cylindrical, nose cap	Level IV	3.2	Gage
110	0.528 diameter, large outside cylindrical, nose cap	Level IV	3.2	Gage
111	0.513 diameter, small outside cylindrical, nose cap	Level IV	3.2	Gage

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TABLE IV. Examination for defects – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Minor				
201	Foreign matter	Level II	3.9	Visual
202	Poor workmanship	Level II	3.9	Visual
Seal, Dwg. 12551778				
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Critical	None defined	N/A	N/A	N/A
Major				
101	0.514 diameter of seal	Level IV	3.2	Gage
102	0.0060 thickness of seal	Level IV	3.2	Gage
Minor				
201	Foreign matter	Level II	3.9	Visual
202	Poor workmanship	Level II	3.9	Visual
Metering Disc, Dwg. 12551779				
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method 3/
Critical	None defined	N/A	N/A	N/A
Major				
101	0.514 outside diameter of metering disc	Level IV	3.2	Gage
102	0.245 inside diameter of metering disc	Level IV	3.2	Gage
103	0.068 thickness of metering disc	Level IV	3.2	Gage
Minor				
201	Foreign matter	Level II	3.9	Visual
202	Poor workmanship	Level II	3.9	Visual

Notes:

1/ A rotating band blank which is deformed or is judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions, shall be classed defective.

2/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicated that the depth or width dimension is borderline.

3/ MIL-STD-651 shall apply in defining and evaluating projectile visual defects.

4.4 Method of inspection.

4.4.1 Banding.

4.4.1.1 Band fill. After sectioning along the longitudinal axis, one half of the sectioned projectile body shall be examined for rotating band fill in the undercut areas. Failure of either of the exposed sections to comply with the 50% fill requirement shall be considered a defect. If any question exists as to the percentage of fill, the sectioned projectile body shall be re-examined under 10x magnification.

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4.4.1.2 Band hardness. The method of test shall be in accordance with ASTM-E18. Failure is defined as any individual hardness number obtained that does not comply with the drawing hardness requirement.

4.4.2 Soundness. The equipment design shall provide controls in accordance with drawing 11075203 that will assure rejection of defective projectile bodies (see 6.9).

4.4.3 Protective coating.

4.4.3.1 Phosphate coating. The method of test shall be as prescribed in TT-C-490 for the type of coating used (see drawing 12551780). The prescribed sample projectiles shall be processed at the start of daily production and at 4 hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.4.3.2 Paint coating. The method of test shall be as prescribed in TT-C-490. The test shall be made on the exterior painted surface of each projectile metal part of the sample forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal on any unit of the sample, as a result of the test, shall be recorded as a defect.

4.4.3.3 Corrosion resistance. The method of test shall be as prescribed in ASTM-B117. The paint shall not be disturbed, however, and unpainted surfaces of the sample projectile metal part may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.5.3.

4.4.4 Metal parts security. The metal parts security shall be in accordance with SCATP-20. The test cartridges, at ambient temperature, shall be fired in bursts of 50 shots in an M39 or M61 gun. The gun barrel shall be at ambient temperature for the firing of each burst. The cyclic rate shall be recorded for informational purposes. Customers, through the procuring activity, may modify the first article test and base acceptance on the firing of 1000 rounds in the M39 or M61 gun.

4.4.5 Crimp ability. A metering disc shall be inserted into each of the sample projectile bodies. The boattail shall be crimped over the metering disc continuously around the periphery of each sample projectile body. The crimped periphery shall be visually examined under 10x magnification for evidence of cracks.

4.4.6 Hardness.

4.4.6.1 Nose cap. The method of test shall be in accordance with ASTM-E18. The hardness of each sample nose shall be measured in areas as shown on drawing 12551773.

4.4.6.2 Projectile body. The projectile body test sample shall be sectioned as shown on drawing 12551782. Care shall be taken not to change the hardness while sectioning the projectile body. Two determinations each shall be made at the indicated positions for each sample body. The method of test for the annealed portion of the projectile body shall be in accordance with ASTM-E384. A minimum indention load of 240 grams shall be used. The method of test for the unannealed portion of the projectile body shall be in accordance with ASTM-E18. Failure is defined as any individual hardness number obtained that does not comply with the drawing hardness pattern position requirement.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor

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personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This item is military unique, and these projectiles are intended for ammunition to be used in 20mm automatic gun systems that have been designed for firing cartridges having the M103 type case configuration.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and all reference documentation cited in this specification (see 2.2.1).
b. Requirements for submission of first article: A first article sample, either in part or complete (Table II), may be required for the commencement of production after the award of a new contract, a change in production venue, a process change for any part or subassembly, or after a production stoppage in excess of 90 days as directed by the government contracting officer.
c. Requirement for submission of inspection equipment designs and manufacturing process.
d. Requirement and provisions for submission of test data as required.
e. Provisions for the inclusion of MIL-STD-1168, Ammunition Data Cards on DD form 1423, Contract Data Requirement List.
f. Provisions for critical characteristic controls.
g. Serialization requirements, if applicable.
h. Critical inspection equipment requirement.
i. Quality Conformance inspection, other than specified in Section 4 of this specification.
j. Applicable National Stock Number.
k. Lists of drawings, publications and specifications, showing applicable revision dates.
l. Certificate of conformance for each lot or shipment of product, if applicable.
m. Place of inspection, if not at place of manufacture.
n. Government Furnished Material or Equipment

6.3 Automatic acceptance inspection equipment (AAIE). Provision concerning the AAIE used to verify the requirements of this specification should be specified in the contract if applicable.

6.4 Submission of inspection equipment designs for approval. Submit copies of designs as required to: Commander, US Army ARDEC, Attn: AMSRD-AAR-QEM-F, Picatinny, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Firing tests. In order to minimize inspection costs, the firing tests will be performed after the sample has been provisionally accepted for all other requirements. Additional cartridges may be required by the test facility (see 4.2). Tests may be performed concurrently on the sample cartridge provided that the test results are not affected by this procedure to minimize testing costs.

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6.6 Test validity. If for any reason the test activity considers that the test conditions have detrimentally affected the test results, the test activity may request the Government to declare the test invalid and authorize a new test.

6.7 Intermediate point inspection. The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality of the end item.

6.8 Malfunction or casualty not covered by this specification. If a lot is suspended due to a malfunction or casualty not covered by this specification, the lot should be referred to the contracting officer.

6.9 Soundness inspection. The contractor should propose a procedure for testing the soundness of projectile bodies that should be approved by the government. Test procedures should be sent to: US Army ARDEC, AMSRD-AAR-QEM-F, Picatinny, NJ 07806-5000.

6.10 Changes from previous issues. Asterisks and marginal notes are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.11 Subject term (key words) listing.

Multipurpose
Multipurpose Ammunition
Incendiary Ammunition
Bursting Charges
Electric Primers
Trace
20 millimeter
Explosives
M940
Ammunition
Nose Fuzes
Projectiles
MPT-SD
Trace Ammunition
Self-Destruct
Self-Destruct Ammunition
Armor Piercing Ammunition

Custodian: Army-AR

Preparing activity: Army-AR
(Project 1305-2006-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil

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